

Global Context Meets Local Quality: A Mamba-Enhanced Detector for Underwater Imagery

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Abstract—Underwater object detection faces severe challenges caused by light attenuation, scattering, spatially varying turbidity, and boundary blur, which weaken object-related visual signals and reduce localization reliability. This letter presents MED, a Mamba-Enhanced Detector for degradation-aware underwater object detection, achieving 82.0% average mAP@0.5 and 48.7% average mAP@0.5:0.95 across three underwater datasets, with 15.5 M parameters and 30.1 GFLOPs. Three problem-driven modules address specific degradation factors: a Mamba-enhanced backbone for selective global-context filtering under attenuated visibility, Dynamic Gated Attention Fusion (DGAF) for adaptive multi-scale feature fusion under non-uniform turbidity, and a Scattering-Aware Quality Estimation (SAQE) head for localization-quality calibration under scattering-induced boundary uncertainty. Extensive experiments on RUOD, UDD, and URPC2020 demonstrate that MED consistently improves detection accuracy over YOLOv11 s and recent underwater detectors, providing a practical accuracy-complexity trade-off for robust underwater perception.

Index Terms—Underwater object detection, State space models, Attention mechanism, Feature fusion, YOLOv11.

I. INTRODUCTION

UNLIKE terrestrial imaging, underwater environments present severe hydro-optical challenges, including wavelength-dependent light absorption, scattering, and non-uniform illumination [1], [2]. These factors lead to image degradation characterized by low contrast, color casting, and blurred object boundaries, making it difficult to distinguish objects from complex backgrounds. Consequently, developing

a robust detector that balances high accuracy with real-time performance for AUV deployment remains an urgent pursuit.

In recent years, Convolutional Neural Networks (CNNs), particularly the single-stage YOLO (You Only Look Once) series, have dominated the field due to their superior inference speed, making them a common baseline for real-time deployment [3]. Nevertheless, underwater targets often exhibit drastic scale changes and degraded textures, and thus many methods introduce feature enhancement and fusion strategies to improve robustness [4], [5], [6], [7], [8], [9], [10], [11], [12]. Meanwhile, Transformer-based underwater detectors improve long-range dependency modeling but are typically heavier in computation [13]. These issues are particularly critical for tiny/blurry targets, where missed detections are common [14].

To overcome the locality bias of CNNs without incurring the quadratic cost of self-attention, Mamba provides linear-time sequence modeling with selective state spaces [15], and VMamba further adapts the SSM paradigm to vision tasks [16]. These developments make it appealing to incorporate Mamba-style modules into underwater detectors. From a signal-processing perspective, the input-dependent selection mechanism is relevant to underwater imagery because object responses are often weakened by attenuation, whereas background responses and ambiguous textures can be amplified by scattering and turbidity. To address these challenges, we develop a Mamba-Enhanced Detector (MED) based on YOLOv11 s. The contributions are threefold: (1) a Mamba-enhanced backbone for selective global-context filtering under attenuation and blur; (2) DGAF for adaptive multi-scale feature fusion under spatially non-uniform turbidity; and (3) SAQE for localization-quality calibration under scattering-induced boundary uncertainty.

II. METHOD

The overall architecture of MED is shown in Fig. 1. It consists of three parts: the Mamba-Enhanced Backbone, the Dynamic Gated Attention Fusion (DGAF) module, and the Scattering-Aware Quality Estimation (SAQE) head.

A. Global Context Modeling Via Mamba-Enhanced Backbone

In underwater imagery, hydro-optical attenuation and scattering make the observed visual signal spatially non-stationary: object boundaries may be weakened in turbid areas, while background texture and suspended particles can introduce ambiguous responses. We therefore use selective state-space modeling to adaptively filter degraded responses while propagating object-related global context. Specifically, we map the 1D input sequence $x(t) \in \mathbb{R}^D$ to a hidden state $h(t) \in \mathbb{R}^N$ through an

Received 13 July 2026; revised 18 August 2026; accepted 22 August 2026. Date of publication 26 August 2026; date of current version 7 September 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62403108 and Grant 42301256, in part by the Fundamental Research Funds for the Central Universities under Grant N25LPY056, and in part by the Science and Technology Planning Project of Liaoning Province under Grant 2023JH1/11200011 and Grant 2024JH2/10240049. The associate editor coordinating the review of this article and approving it for publication was Dr. Raghu G. Raj. (Yaoming Zhuang and Zirui Fang are co-first authors.) (Corresponding authors: Yaoming Zhuang; Qiong Zhou.)

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Digital Object Identifier 10.1109/LSP.2026.3727946

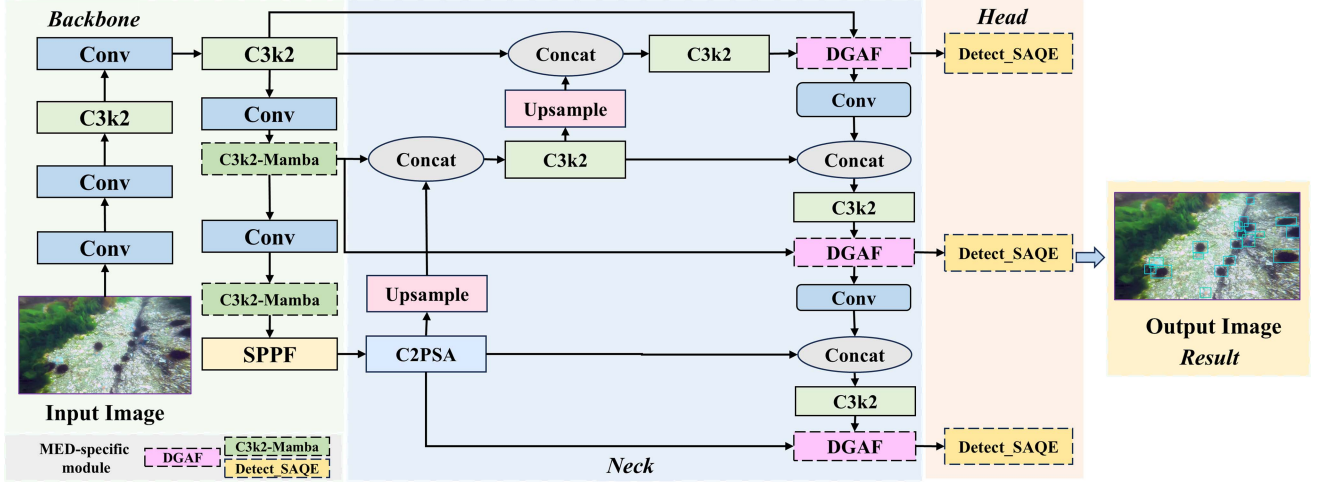


Fig. 1. Framework of the proposed MED. The input underwater images are enhanced and feature extraction is performed on the main trunk. Then, multi-scale dynamic fusion is carried out through the DGAF module, and finally processed by the SAE head to obtain the detection results.

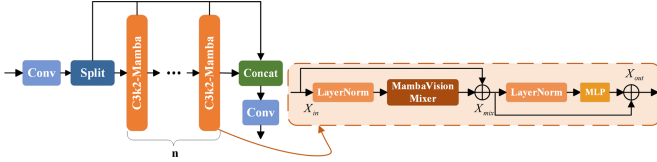


Fig. 2. Framework of the proposed Global Context Modeling via Mamba-Enhanced Backbone.

LTI system and discretize it with ZOH:

$$\begin{aligned}\bar{\mathbf{A}} &= \exp(\Delta \mathbf{A}), \\ \bar{\mathbf{B}} &= (\Delta \mathbf{A})^{-1} (\exp(\Delta \mathbf{A}) - \mathbf{I}) \Delta \mathbf{B}, \\ \mathbf{h}_t &= \bar{\mathbf{A}} \mathbf{h}_{t-1} + \bar{\mathbf{B}} \mathbf{x}_t, \quad \mathbf{y}_t = \mathbf{C} \mathbf{h}_t \cdot \text{SiLU}(\mathbf{z}_t).\end{aligned}\quad (1)$$

where $\mathbf{A}$ represents the structured state matrix, and $\mathbf{B}, \mathbf{C}$ are projection parameters. A critical property in this design is the selection mechanism, where the parameters $\Delta, \mathbf{B}, \mathbf{C}$ are derived from the input $\mathbf{x}_t$ rather than being time-invariant. This input-dependent selectivity enables the model to dynamically attenuate degraded background responses caused by hydro-optical scattering while propagating object-related structural cues over a long range [16]. In our YOLOv11s-based implementation, the Mamba-enhanced blocks are inserted only in the high-level backbone, after the P4/16 and P5/32 downsampling stages, corresponding to layers 6 and 8 in the model configuration. No Mamba-enhanced block is used in the shallow P1–P3 stages or in the detection head.

Specific module details are illustrated in Fig. 2. Within each Mamba-enhanced block, we adopt a Sequential Mamba-Attention Aggregation strategy. The feature definition X_{in} is first processed by the SSM module to capture global semantic context, followed by a multi-head self-attention mechanism to refine local discriminative details. This process can be mathematically expressed as:

$$X_{mix} = \text{MambaVisionMixer}(\text{LN}(X_{in})) + X_{in}, \quad (2)$$

$$X_{out} = \text{MLP}(\text{LN}(X_{mix})) + X_{mix}, \quad (3)$$

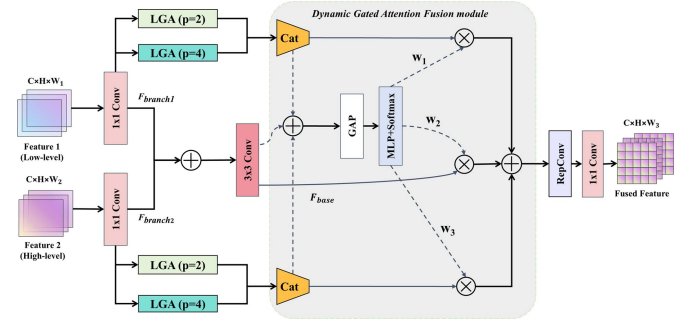


Fig. 3. The network architecture of Dynamic Gated Attention Fusion Mechanism, where the solid lines represent the flow of feature maps and the dashed lines represent the flow of weights.

where $\text{LN}(\cdot)$ denotes Layer Normalization. This sequence combines state-space global filtering with attention-based feature refinement, while the early CNN layers retain high-frequency texture cues.

B. Dynamic Gated Attention Fusion Mechanism

Underwater degradation is spatially heterogeneous, so shallow boundary cues and deeper semantic cues may have different reliability across regions. DGAF replaces static aggregation with content-aware multi-scale gating and is applied at the P3, P4, and P5 detection scales, with layer pairs [4,16], [6,19], and [10,22], respectively. Specific module details are illustrated in Fig. 3.

Formally, given input feature maps X_{low}, X_{high} from adjacent pyramid levels, we first construct a multi-branch topology to extract diverse representations. Each input is projected to half of the output channels by a 1×1 convolution. A structural branch generates a baseline feature map F_{base} through grouped 3×3 convolution on the summed projected features to preserve spatial geometry. Simultaneously, two contextual branches use pooling scales $p = 2$ and $p = 4$ to capture local-context and global-context cues, yielding multi-granular semantic descriptors F_{attn1} and F_{attn2} .

The adaptive selection in DGAF is inspired by the selective kernel mechanism [17]. A lightweight Multi-Layer Perceptron

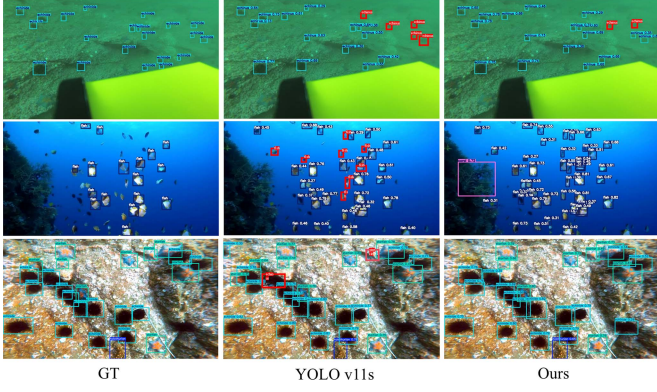


Fig. 4. Visual comparison results of GT with YOLOv11s and Ours on RUOD or UDD Datasets. The red boxes indicate the objects that the model failed to detect.

(MLP) maps the global descriptor $\mathbf{s}$ to channel-wise attention vectors. A Softmax operator is then applied across the branches to normalize these weights, ensuring a probabilistic distribution. Let $\alpha_c, \beta_c, \gamma_c$ denote the learned gating weights for the three branches at the c -th channel. The final fused feature map $\mathbf{V}_c$ is derived through a weighted aggregation:

$$\begin{aligned} \mathbf{V}_c &= \alpha_c \cdot \mathbf{F}_{\text{base},c} + \beta_c \cdot \mathbf{F}_{\text{attn1},c} + \gamma_c \cdot \mathbf{F}_{\text{attn2},c}, \\ \text{s.t. } \alpha_c + \beta_c + \gamma_c &= 1. \end{aligned} \quad (4)$$

Finally, the branch outputs are concatenated and refined by a 1×1 convolution, a Re-parameterization Convolution (Rep-Conv) [18], and a final 1×1 convolution.

C. Scattering-Aware Quality Estimation Head

Although anchor-free detectors can model regression uncertainty [36], top- k flatness alone may not distinguish background noise from blurry underwater targets. To bridge this gap, we propose the Scattering-Aware Quality Estimation (SAQE) head. SAQE combines entropy and local distribution peaks to construct a localization-quality descriptor and calibrate detection confidence under scattering-induced boundary ambiguity.

Formally, let $\mathbf{P} \in \mathbb{R}^n$ denote the predicted probability distribution over the discrete regression bins for a bounding box edge, where n is the number of bins. We first extract the top- k probabilities $\mathcal{T} = \text{Top}_k(\mathbf{P})$ and their mean value $\mu_{\mathcal{T}}$ to capture the primary mode of the distribution. To explicitly measure the dispersion caused by water turbidity, we compute the Shannon Entropy $\mathcal{H}(\mathbf{P})$:

$$\mathcal{H}(\mathbf{P}) = - \sum_{i=1}^n p_i \log(p_i + \epsilon), \quad (5)$$

where p_i represents the probability at the i -th bin and ϵ is a small constant for numerical stability. A higher $\mathcal{H}(\mathbf{P})$ indicates a flatter distribution, corresponding to severe scattering or occlusion, whereas a lower $\mathcal{H}(\mathbf{P})$ implies a sharp, unambiguous boundary [37]. In our implementation, SAQE uses top- $k = 4$ and $\text{reg_max} = 16$. For each of the four bounding-box sides, the top-four probabilities and their mean response are concatenated, resulting in 20 quality-descriptor channels. The quality-estimation MLP has hidden dimension 64 and structure $20 \rightarrow 64 \rightarrow 1$, implemented with 1×1 convolutional layers and

TABLE I
ABLATION STUDY ON THREE UNDERWATER DATASETS

Backbone	DGAF	SAQE	P(M)	G	FPS	UDD	URPC2020	RUOD
—	—	—	9.41	21.3	270	67.9 ± 0.4	83.9 ± 2.3	85.0 ± 0.6
✓	—	—	13.86	25.1	—	27.9 ± 0.4	49.6 ± 2.5	62.0 ± 0.2
—	✓	—	11.53	23.7	—	$68.9 / 29.3$	$85.4 / 51.1$	$86.1 / 63.7$
—	—	✓	9.76	21.6	—	$69.6 / 29.1$	$86.1 / 51.3$	$86.4 / 64.1$
—	—	—	15.50	30.1	286	$69.8 / 29.9$	$85.9 / 50.9$	$86.6 / 63.5$
✓	✓	✓	15.50	30.1	286	72.4 ± 0.4	86.7 ± 1.0	87.2 ± 0.4
						31.0 ± 0.1	51.2 ± 1.2	64.2 ± 0.1

Note: Dataset cells list mAP@0.5 (top) and mAP@0.5:0.95 (bottom).

Baseline/complete MED: mean $\pm$ 95% CI over three seeds (two-sided Student's t). P(M): million parameters; G: GFLOPs.

ReLU activation. The predicted quality score is used to calibrate the classification confidence before final prediction.

The descriptor $\mathbf{S} = [\mathcal{T}, \mu_{\mathcal{T}}, \mathcal{H}(\mathbf{P})]$ is fed into an MLP Φ to refine the classification score:

$$C_{\text{final}} = C_{\text{cls}} + \Phi(\mathbf{S}), \quad \text{where } \Phi(\mathbf{S}) \in \mathbb{R}. \quad (6)$$

C_{cls} denotes the original classification score, $\Phi(\mathbf{S})$ is the learned scalar localization-quality offset produced from the descriptor $\mathbf{S}$, and C_{final} denotes the calibrated score used for final prediction.

III. EXPERIMENTS

A. Datasets

We conduct experiments on three challenging underwater benchmarks: RUOD [38], UDD [39], and URPC2020 [40]. RUOD contains 14,000 images with 74,903 instances from 10 categories, split into 9,800/4,200 images for training/validation. UDD includes 2,227 images from three benthic categories, with 1,558/335/334 images for training/validation/test. For URPC2020, we use 4,434 images for training and 1,109 images for validation.

B. Implementation Details

All experiments were conducted on Ubuntu 22.04 LTS with an NVIDIA GeForce RTX 3090 GPU (24 GB), PyTorch 2.1.2, CUDA 12.1, and an input resolution of 640×640 . The models were trained for 200 epochs without pretrained weights, using a momentum of 0.937, a learning rate of 0.01, and a batch size of 16. Evaluation metrics include Precision (P), Recall (R), mAP@0.5, mAP@0.5:0.95, parameters, and GFLOPs. The FPS of YOLOv11s and MED was measured under FP32 PyTorch inference with a batch size of 16.

C. Ablation Experiment

The ablation results in Table I show that MED consistently improves mAP@0.5/mAP@0.5:0.95 over YOLOv11s, with gains of 4.4/2.9 points on UDD, 1.7/0.2 points on URPC2020, and 2.1/2.2 points on RUOD. Among the individual modules, the Mamba-enhanced backbone improves long-range global context modeling, DGAF strengthens adaptive multi-scale fusion, and SAQE benefits localization-quality estimation under boundary uncertainty. Their combination achieves the best overall results on all three datasets, indicating complementary gains under different underwater degradation conditions.

TABLE II
COMPREHENSIVE PERFORMANCE COMPARISON ON THE UDD AND URPC DATASETS

Model	Source	P(M)	G	Precision (%)			Recall (%)			mAP@0.5 (%)			mAP@0.5:0.95 (%)		
				UDD	URPC2020	AVG	UDD	URPC2020	AVG	UDD	URPC2020	AVG	UDD	URPC2020	AVG
YOLOv6s [19]	arXiv22	16.29	43.7	65.9	82.4	74.2	59.6	77.8	68.7	65.5	83.9	74.7	28.4	49.2	38.8
YOLOv7s [20]	CVPR23	9.33	26.0	64.9	82.2	73.6	56.8	75.4	66.1	62.0	82.2	72.1	24.1	44.3	34.2
YOLOv8s [21]	Ultralytics	11.13	28.4	64.8	84.6	74.7	60.1	77.1	68.6	65.8	84.6	75.2	28.4	50.1	39.2
YOLOv10s [3]	NeurIPS24	8.04	24.5	72.5	82.2	77.3	59.6	78.3	69.0	68.4	84.5	76.5	28.8	49.6	39.2
YOLOv12s [22]	NeurIPS25	9.07	19.3	69.0	84.0	76.5	62.6	76.2	69.4	67.8	85.0	76.4	29.2	50.4	39.8
Hyper-YOLO-s [23]	TPAMI25	13.52	33.8	75.2	84.2	79.7	62.0	78.0	70.0	67.0	84.8	75.9	29.0	50.6	39.8
Mamba-YOLO-B [24]	AAAI25	21.80	49.6	75.8	82.3	79.0	57.5	79.4	68.5	67.2	84.6	75.9	29.0	50.1	39.5
RT-DETR(Res50) [25]	CVPR24	32.66	85.1	62.6	82.1	72.3	54.1	78.4	66.2	59.2	84.2	71.7	26.2	48.2	37.2
Dynamic-YOLO [26]	AI Rev24	8.21	29.4	76.8	82.9	79.8	54.5	76.9	65.7	67.6	84.4	76.0	28.2	50.0	39.1
VajraV1-small [27]	arXiv25	11.41	–	72.7	81.3	77.0	59.2	77.1	68.2	66.6	83.0	74.8	28.7	48.5	38.6
U-DECN [28]	TGRS25	82.00	170	70.6	82.8	76.7	53.1	78.1	65.6	59.4	82.1	70.8	23.1	42.2	32.7
LUOD-YOLO-s [29]	JRTIP25	5.91	19.9	65.8	82.7	74.2	62.7	78.8	70.8	64.8	84.3	74.5	28.1	49.5	38.8
SU-YOLO [30]	OE23	6.97	–	65.8	77.4	71.6	61.1	72.0	66.5	63.3	76.8	70.0	26.2	39.3	32.8
EAST-YOLO [9]	SPL25	2.69	6.8	67.3	81.9	74.6	64.7	76.0	70.3	67.9	82.5	75.2	28.2	47.7	38.0
YOLOv11s [31]	Ultralytics	9.41	21.3	66.4	83.6	75.0	69.3	77.3	73.3	68.0	84.9	76.5	28.1	50.8	39.5
Ours	–	15.50	30.1	76.9	83.2	80.1	62.9	80.5	71.7	72.4	86.6	79.5	31.0	51.0	41.0

Note: Blue shades denote the best, second-best, and third-best results in each metric column, from dark to light.

TABLE III
COMPARATIVE RESULTS ON THE RUOD DATASET

Model	Source	P(M)	G	AP@0.5 / mAP (%)											
				holothurian	echinus	scallop	starfish	fish	corals	diver	cuttlefish	turtle	jellyfish	mAP@0.5 (%)	mAP@0.5:0.95 (%)
YOLOv5s [32]	Ultralytics	7.03	16.5	80.7	90.5	82.4	89.8	75.7	74.9	93.8	97.0	97.0	77.0	85.9	62.1
YOLOv6s [19]	arXiv22	16.29	43.7	81.3	90.5	80.1	89.6	74.7	73.6	93.1	97.0	97.1	74.7	85.2	62.2
YOLOv8s [21]	Ultralytics	11.13	28.4	81.8	91.6	82.8	90.8	75.8	74.5	94.2	97.2	96.8	77.4	86.3	62.9
YOLOv9s [33]	CVPR24	7.17	26.7	82.6	90.7	82.7	90.5	76.0	74.8	94.3	97.3	97.0	77.8	86.4	63.9
YOLOv10s [3]	NeurIPS24	8.04	24.5	81.3	91.8	82.7	90.5	74.9	73.7	93.2	97.3	96.8	80.0	86.2	63.3
YOLOv12s [22]	NeurIPS25	9.07	19.3	82.0	91.5	82.0	90.5	76.2	75.3	94.5	97.1	97.1	78.4	86.5	63.9
YOLOv13s [34]	arXiv25	9.00	20.7	80.6	90.5	82.2	90.2	74.4	74.9	94.1	96.7	97.1	78.9	85.9	62.9
YOLOv26s [35]	Ultralytics	9.77	27.8	81.1	90.6	82.6	89.9	75.4	75.2	94.2	96.8	96.9	79.4	86.2	63.6
Mamba-YOLO-B [24]	AAAI25	21.80	49.6	82.4	91.0	81.7	90.9	76.9	74.9	94.3	97.2	97.1	79.2	86.3	63.6
RT-DETR(Res50) [25]	CVPR24	32.66	85.1	82.1	90.3	81.7	89.5	75.2	72.2	92.8	97.0	96.1	77.2	85.4	61.1
EAST-YOLO [9]	SPL25	2.69	6.8	79.0	90.7	77.0	89.1	72.2	72.3	92.6	96.8	96.4	75.6	84.2	60.4
YOLOv11s [31]	Ultralytics	9.41	21.3	81.3	89.7	80.2	89.4	73.2	74.0	93.6	96.9	96.8	75.0	85.0	62.0
Ours	–	15.50	30.1	82.6	91.8	81.1	91.0	78.2	75.5	94.6	97.1	97.3	81.2	87.1	64.2

Note: Blue shades denote the best, second-best, and third-best results in each metric column, from dark to light.

D. Comparative Experiment

We compare MED with representative YOLO variants, Transformer-based detectors, and recent underwater detectors [3], [19], [21], [23], [24], [25], [30], [32], [33]. The results in Table II show that MED achieves the best average Precision, mAP@0.5, and mAP@0.5:0.95 on UDD and URPC2020, with average mAP gains of 3.0 and 1.5 points over YOLOv11s, respectively. These improvements indicate that MED is effective for both low-contrast benthic scenes and multi-object underwater scenarios. On the large-scale, long-tailed, multi-class RUOD dataset, Table III shows that MED still obtains strong overall performance, achieving the best mAP@0.5/mAP@0.5:0.95 of 87.1%/64.2% and top-ranked AP on most object categories. Although MED is not the lightest model, its 15.50 M parameters and 30.1 GFLOPs are lower than those of Mamba-YOLO-B, RT-DETR(Res50), and U-DECN, supporting a favorable accuracy-complexity trade-off for robust underwater deployment.

IV. CONCLUSION

This letter presents MED, a degradation-aware underwater detection architecture that integrates selective state-space

modeling, dynamic multi-scale feature fusion, and localization-quality calibration. With 15.50 M parameters and 30.1 GFLOPs, MED achieves 82.0% average mAP@0.5 and 48.7% average mAP@0.5:0.95 across RUOD, UDD, and URPC2020, improving YOLOv11s by 2.7 and 1.7 percentage points, respectively. The results demonstrate that targeted adaptations to underwater signal degradation can improve detection robustness under attenuation, non-uniform turbidity, and scattering-induced boundary uncertainty. The complementary roles of the Mamba-enhanced backbone, DGAF, and SAQE contribute to both accuracy and reliability, while maintaining a practical accuracy-complexity trade-off for underwater perception. Future work will investigate how to deploy faster, achieve greater robustness under severe turbidity and lighting variations, and generalize across datasets.

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